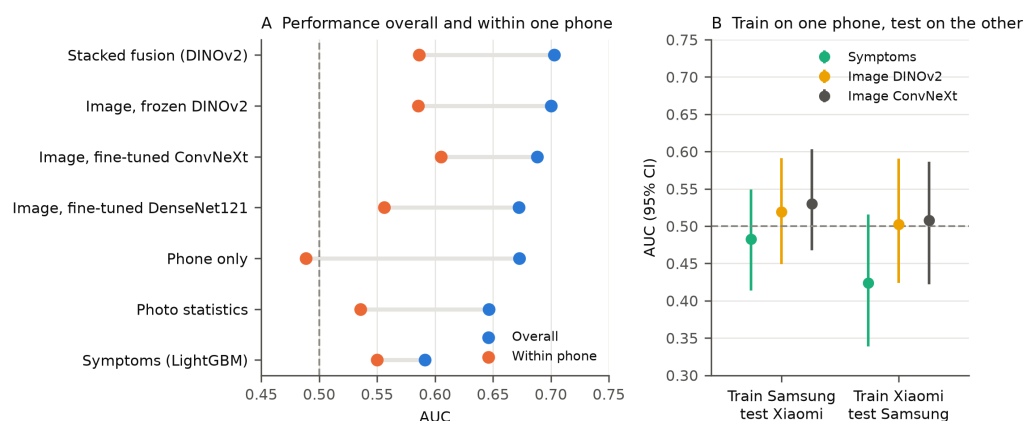


# Graphical Abstract

## Device and site shortcuts dominate smartphone throat-image classification of bacterial pharyngitis: a multimodal benchmark on PGUPharyngitis

Sandesh Bhatta



## Highlights

### **Device and site shortcuts dominate smartphone throat-image classification of bacterial pharyngitis: a multimodal benchmark on PGUPharyngitis**

Sandesh Bhatta

- First multimodal benchmark on PGUPharyngitis: images, 20 symptoms, age and sex.
- Phone model alone predicts expert labels (AUC 0.67), beating the published baseline.
- Image features identify the phone (AUC 0.99) but fail across phones (AUC  $\approx$  0.5).
- Within one phone, image AUC falls to 0.50–0.58; 18 colour statistics match deep nets.
- Symptom fusion adds nothing over images; benchmarks need site-aware evaluation.

# Device and site shortcuts dominate smartphone throat-image classification of bacterial pharyngitis: a multimodal benchmark on PGUPharyngitis

Sandesh Bhatta<sup>a,\*</sup>

<sup>a</sup>*Department of Computer Science, University at Albany, State University of New York, Albany, New York, USA*

---

## Abstract

*Background:* Smartphone throat photographs could help separate bacterial from non-bacterial pharyngitis. The public PGUPharyngitis dataset (742 patients, two cities, two phones) reports image-only AUCs up to 0.645, but no study has added symptoms or checked what the models learn.

*Methods:* We analysed 735 patients labelled by majority vote of 4–9 physicians, using 5-fold cross-validation repeated five times with nested tuning. We compared symptom models, frozen and fine-tuned image networks, standard fusion and a gated fusion network. We then tested for device shortcuts with phone-only and photo-statistics baselines, phone identification, within-phone and cross-phone evaluation, and three rules for tied votes.

*Results:* Stacked image–symptom fusion reached an AUC of 0.703 (95% CI 0.66–0.74), images alone 0.700 and symptoms alone 0.596. Knowing only the phone gave 0.673, and 18 colour statistics gave 0.646, matching the published baseline. Image features identified the phone with an AUC of 0.99. Within one phone, image AUC fell to 0.59, and models trained on one phone scored 0.50–0.53 on the other. Fusion beat symptoms under one of three tie rules and never beat images alone.

*Conclusions:* Most of the apparent performance on PGUPharyngitis comes from recognising the phone and site that took the photograph, not the throat. Smartphone pharyngitis models need within-site and cross-site evaluation be-

---

\*Corresponding author.

Email address: [sbhatta@albany.edu](mailto:sbhatta@albany.edu) (Sandesh Bhatta)

URL: <https://orcid.org/0009-0008-0399-0480> (Sandesh Bhatta)

fore any clinical use.

*Keywords:* pharyngitis, smartphone imaging, shortcut learning, multimodal fusion, deep learning, antibiotic stewardship

---

## 1. Introduction

Sore throat is one of the most common reasons for primary-care visits, and most cases are viral. Group A streptococcus and other bacteria cause a minority of cases, yet antibiotics are prescribed far more often than bacterial infection occurs [1, 2]. Clinical scores such as Centor and McIsaac combine fever, tonsillar exudate, tender cervical lymph nodes, absence of cough and age to estimate the probability of streptococcal infection [3, 4]. They are simple and widely taught, but their discrimination is moderate and they depend on a careful physical examination.

Smartphone photographs of the throat are a cheap alternative that also works remotely. Deep-learning models trained on such photographs have detected severe pharyngitis [5] and streptococcal pharyngitis [6] in small single-site studies. In 2025, Shojaei et al. released PGUPharyngitis, the largest public dataset of its kind: 742 throat photographs from patients with cold symptoms in two Iranian cities, taken with two phone models, each labelled bacterial or non-bacterial by four to nine physicians, together with 20 symptoms, age and sex [7]. Their image-only baselines reached an AUC of 0.645 (DenseNet121), and the dataset was offered to support AI-based diagnosis and remote care.

Two questions remain open. First, the dataset records symptoms, and physicians labelled each case from the photograph and the symptoms together, yet no published model combines the two. Multimodal fusion of images with clinical variables often improves medical prediction [8], so fusion is worth testing here. Second, nobody has checked what image models learn on this dataset. Deep networks readily exploit “shortcuts”: features that predict the label in the training data but are unrelated to the disease [9]. In chest radiography, models identified the hospital and scanner and used them to predict pneumonia or COVID-19 [10, 11], and hip-fracture models leaned on scanner and patient variables [12]. In PGUPharyngitis, each city used a different phone, and bacterial prevalence may differ between cities. A model could therefore score well by recognising the phone.

We set out to build a symptom-aware image model and, in doing so, to test for device and site shortcuts. Our contributions are:

1. The first multimodal benchmark on PGUPharyngitis, comparing symptom-only, image-only, clinical-score and five fusion approaches, including a gated cross-modal fusion network, under one repeated, nested cross-validation protocol with public folds.
2. A shortcut analysis showing that the phone alone, or 18 global colour statistics, match the published image baseline; that image features identify the phone almost perfectly; and that image performance largely disappears within a phone and across phones.
3. Ablation, label-dependence and tie-rule sensitivity analyses showing which conclusions hold under different analysis choices and which do not.
4. Open code, fold assignments and out-of-fold predictions to make future work on this dataset comparable.

## 2. Related work

*Clinical scores.* The Centor score assigns one point each for fever, tonsillar exudate, tender anterior cervical lymphadenopathy and absence of cough [3]; McIsaac added an age adjustment [4]. Both are used to decide on testing or empirical antibiotics, and both need findings from a physical examination. PGUPharyngitis records fever and cough but not exudate or lymph nodes, so only a modified score can be computed.

*Throat-image AI.* Yoo et al. trained convolutional networks on smartphone throat photographs to separate severe pharyngitis from normal throats [5], and Askarian et al. used colour and texture features from smartphone images to detect streptococcal pharyngitis [6]. These studies were small and from single sites. Shojaei et al. released PGUPharyngitis with image-only baselines evaluated by three-fold cross-validation, reporting AUCs from 0.554 (MobileNetV3) to 0.645 (DenseNet121) on the full dataset [7]. To our knowledge, no study has fused these images with symptoms or examined what the models learn.

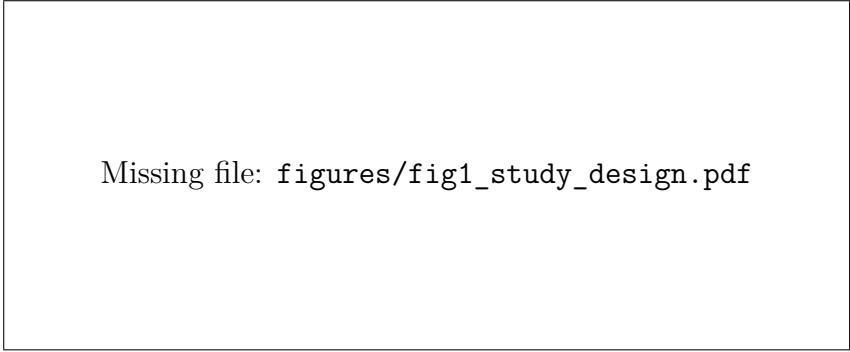
*Multimodal fusion.* Medical image–tabular fusion is usually grouped into early fusion (concatenating features), late fusion (combining predictions) and joint or intermediate fusion (learning a shared representation) [8]. Gated

66 units learn how much to trust each modality per sample [13], and feature-  
67 wise linear modulation (FiLM) lets one modality scale and shift the features  
68 of another [14]. We implement all of these families on equal terms.

69 *Shortcut learning and site confounding.* Shortcut learning occurs when a  
70 model relies on cues that are predictive in the training distribution but not  
71 causally related to the target [9]. In medical imaging, such cues include  
72 the hospital, scanner, positioning markers and patient metadata [10, 11, 12].  
73 Hidden stratification, where performance is driven by an easy subgroup, can  
74 make overall metrics misleading [15]. Methodological reviews recommend  
75 evaluation across sites, explicit confounder baselines and reporting of variance  
76 across splits [16]. We apply these recommendations to throat imaging.

### 77 3. Materials and methods

78 The study is reported following TRIPOD+AI [17]. All analyses are ex-  
79 ploratory: the analysis plan was written before modelling, but hypotheses  
80 were not formally frozen before the first results were seen. Figure 1 sum-  
81 marises the study design.



Missing file: figures/fig1\_study\_design.pdf

Figure 1: Study design: 742 patients, 735 analysed; inputs (photograph, 20 symptoms, age, sex), model families,  $5 \times 5$  nested cross-validation and the shortcut analyses.

#### 82 3.1. Dataset

83 We used PGUPharyngitis (Figshare, <https://doi.org/10.6084/m9>  
84 [.figshare.28163513](https://doi.org/10.6084/m9.figshare.28163513), CC BY 4.0) [7]. It contains one throat photo-  
85 graph per patient, taken with a Samsung Galaxy S21 Ultra or a Xiaomi  
86 phone in two Iranian cities between October 2023 and May 2024, and a

table with age, sex, 20 binary symptoms and four to nine physician diagnoses (bacterial or non-bacterial). The original study received ethics approval (IR.BPUMS.REC.1403.282) with written informed consent; we used the anonymised public release only.

The table lists 742 patients. Patient 666 had no image. Three pairs of patients (19/23, 218/256, 301/305) shared an identical photograph (perceptual-hash distance 0) but had different ages, sexes or symptoms; because the image could not be assigned to either record, we excluded all six. The analysis set has 735 patients. The table has no city or phone column, so we read the phone model from each photograph’s EXIF metadata: Samsung (model code SM-G998B) for 382 patients, Xiaomi (2201117SG) for 290, and unknown for 63. Because each city used one phone [7], we treat the phone as a proxy for acquisition site and write “phone/site” throughout.

### 3.2. Labels

The 735 patients received 3,662 diagnoses (median 5 per patient; one patient had 3 recorded). The reference label is the majority vote. One hundred patients (13.6%) had a tied vote; in the primary analysis ties count as bacterial, giving 205 bacterial cases (27.9%). Two sensitivity analyses count ties as non-bacterial (105 bacterial, 14.3%) or exclude them (635 patients, 16.5% bacterial). We also keep the soft label, the share of physicians voting bacterial, and the agreement,  $|\text{soft} - 0.5| \times 2$ ; 230 patients (31%) had unanimous votes. The labels reflect expert consensus from photographs and symptoms, not microbiological testing.

### 3.3. Preprocessing

Images were rotated according to EXIF orientation, converted to RGB and resized so the short side was 256 pixels. Training augmentation used a random resized crop to  $224 \times 224$  (scale 0.8–1.0), horizontal flip, rotation up to  $15^\circ$  and brightness and contrast jitter of 10%. Hue and saturation were not altered, because throat redness is diagnostic. At test time we used a 224-pixel centre crop and averaged predictions over the original and a horizontally flipped image. Images were normalised with ImageNet statistics. Tabular inputs were the 20 symptoms, sex (0/1) and age; age was standardised and any missing value imputed using statistics from the training data only (no values were missing).

### 121 3.4. Evaluation protocol

122 All models used the same stratified 5-fold cross-validation repeated five  
123 times with different seeds (25 test folds of 147 patients), stratified on label  
124  $\times$  phone. Fold assignments are released with the code. Within each training  
125 part, a 3-fold inner cross-validation selected hyperparameters (by AUC), the  
126 number of training epochs for neural fusion models, and the decision thresh-  
127 old, chosen to maximise balanced accuracy on inner out-of-fold predictions.  
128 Scalers, imputers, PCA, thresholds and stacking meta-models were fitted on  
129 training data only. Because fine-tuning is costly, fine-tuned CNNs used one  
130 repeat (5 folds) and a stratified 15% validation split of each training part for  
131 early stopping and threshold choice.

### 132 3.5. Models

133 *Symptom models.* Logistic regression (L2, class-balanced,  $C \in \{0.01, 0.1, 1, 10\}$ ),  
134 LightGBM [18] (4 or 8 leaves, 100 or 300 trees, learning rate 0.05) and a  
135 two-layer perceptron (64–64 units, L2 penalty  $\in \{0.001, 0.1, 1\}$ ) used the 20  
136 symptoms, sex and age.

137 *Clinical score.* A modified McIsaac score summed fever (“Pyrexia”), absence  
138 of cough and an age term (+1 for 3–14 years,  $-1$  for  $\geq 45$  years) [4]; exudate  
139 and lymph-node items are not recorded.

140 *Reference baselines.* To measure shortcuts we trained three models that do  
141 not use throat anatomy: (i) *phone only*, predicting the training-fold bacterial  
142 rate of the photograph’s phone; (ii) *image size*, logistic regression on the log  
143 width, height, aspect ratio and area of the original photograph; and (iii) *photo*  
144 *statistics*, logistic regression on 18 global statistics of each resized photograph  
145 (RGB and HSV means and standard deviations, shares of dark, bright and  
146 strongly red pixels, and edge density).

147 *Frozen image models.* We extracted embeddings from DINOv2 ViT-S/14  
148 [19] (384 dimensions, 224-pixel input) and ConvNeXt-Tiny pretrained on  
149 ImageNet-22k and fine-tuned on ImageNet-1k [20] (768 dimensions), once per  
150 image for the test view, the flipped view and ten random training augmen-  
151 tations. Standardised embeddings were classified by class-balanced logistic  
152 regression ( $C \in \{0.001, 0.01, 0.1, 1\}$ ).

153 *Fine-tuned image models.* To reproduce the published baselines we fine-tuned  
 154 DenseNet121 [21], MobileNetV3-Large [22] and ConvNeXt-Tiny end to end  
 155 with AdamW (learning rate  $10^{-4}$ , weight decay  $10^{-4}$ , cosine schedule, batch  
 156 16), class-weighted cross-entropy, up to 30 epochs and early stopping on  
 157 validation AUC (patience 7).

158 *Standard fusion.* *Late fusion* averaged the symptom and image logistic-regression  
 159 probabilities. *Early fusion* fitted one logistic regression on the tabular inputs  
 160 plus a PCA projection of the image embedding (16 or 64 components). *Stack-*  
 161 *ing* fitted a logistic regression on the logits of inner out-of-fold symptom and  
 162 image predictions.

163 *Gated cross-modal fusion.* The image embedding passes through layer nor-  
 164 malisation, dropout and a linear layer to a 128-dimensional vector  $h_{\text{img}}$ ; the  
 165 22 tabular inputs pass through a 22–64–128 perceptron to  $h_{\text{sym}}$ . A gate  
 166 computes a scalar weight per patient,

$$g = \sigma(W[h_{\text{img}}; h_{\text{sym}}] + b), \quad z = g h_{\text{img}} + (1 - g) h_{\text{sym}}, \quad (1)$$

167 and a dropout–linear head maps  $z$  to a logit. Auxiliary image-only and  
 168 symptom-only heads add their losses with weight  $\lambda$ , and modality dropout  
 169 (probability 0.2 per branch, never both) randomly removes one branch during  
 170 training. Training used the soft label as target in class-weighted binary cross-  
 171 entropy, AdamW with weight decay  $10^{-4}$ , batch 32 and a cosine schedule over  
 172 60 epochs. Each epoch drew one of the ten cached augmentation embeddings  
 173 per patient. The inner cross-validation selected the learning rate  $\{10^{-3}, 3 \times$   
 174  $10^{-4}\}$ , dropout  $\{0.3, 0.5\}$ ,  $\lambda \in \{0, 0.3\}$  and the number of epochs (maximum  
 175 of the smoothed mean inner AUC curve); the model was then retrained on  
 176 the full training part. Ablations replaced the gate by concatenation or by  
 177 FiLM [14], used hard labels, removed the auxiliary heads, removed modality  
 178 dropout, or removed fever and cough from the inputs.

### 179 3.6. Shortcut and label-dependence analyses

- 180 1. *Phone identification:* logistic regression predicting Xiaomi versus Sam-  
 181 sung (672 patients with known phone) from image embeddings, image  
 182 size or photo statistics, in 5-fold cross-validation.
- 183 2. *Within-phone evaluation:* for each model, AUC computed separately  
 184 within the Samsung, Xiaomi and unknown-phone strata of the out-of-  
 185 fold predictions and averaged weighted by stratum size. A phone-only  
 186 model scores 0.5 by construction.

- 187 3. *Within-phone training*: models trained and tested inside one phone  
188 (5-fold  $\times$  5 repeats).
- 189 4. *Cross-phone transfer*: models trained on all patients of one phone and  
190 tested on the other, with patient-level bootstrap confidence intervals  
191 (1,000 resamples).
- 192 5. *Large images only*: within-phone AUC restricted to photographs whose  
193 original short side was at least 500 pixels.
- 194 6. *Shuffled-image control*: gated fusion trained after permuting image em-  
195 beddings across patients; with no leakage it should lose the image sig-  
196 nal.
- 197 7. *Label dependence*: AUC on unanimous versus split votes; Spearman  
198 correlation with the vote share; and a residual test in which DINOv2  
199 embeddings predict the out-of-fold errors of the symptom model (ridge  
200 regression, 5-fold cross-validated correlation, 500 permutations).

### 201 3.7. Statistics

202 For each model and repeat, the five test folds were pooled and AUC, PR-  
203 AUC, balanced accuracy, macro F1, sensitivity, specificity, Brier score and  
204 calibration slope were computed; we report the mean  $\pm$  SD across repeats.  
205 AUC 95% confidence intervals come from 1,000 patient-level bootstrap re-  
206 samples within each repeat, averaged across repeats. Model comparisons  
207 used the Nadeau–Bengio corrected resampled  $t$ -test on the 25 fold-level AUC  
208 differences [23], which accounts for overlapping training sets, and the De-  
209 Long test [24, 25] within each repeat. Three hypotheses compared gated  
210 fusion with the best symptom-only model (H1), the best image-only model  
211 (H2) and the best standard fusion model (H3), one-sided, with Holm cor-  
212 rection [26]; choosing the best comparator by its own performance makes  
213 these tests conservative. Decision curves [27] show net benefit for threshold  
214 probabilities 0.05–0.60.

### 215 3.8. Explainability and implementation

216 We computed SHAP values [28] for a LightGBM symptom model, sum-  
217 marised gate values by vote agreement, and produced Grad-CAM maps [29]  
218 from a ConvNeXt-Tiny fine-tuned on all patients for the median best epoch  
219 count (seven), used only for visualisation. Code ran on an Apple M5 Mac-  
220 Book Air (16 GB) with Python 3.11, PyTorch 2.14 (Metal backend), timm  
221 1.0, scikit-learn 1.9 and LightGBM 4.7.

## 222 4. Results

### 223 4.1. Cohort

224 The 735 patients had a mean age of 28.8 years (SD 17.1, range 2–83)  
 225 and 373 (50.7%) were male (Table 1). Bacterial prevalence differed threefold  
 226 between phones: 14.9% for Samsung photographs and 43.8% for Xiaomi  
 227 photographs.

Table 1: Cohort characteristics (primary analysis, ties counted as bacterial).

| Characteristic                                  | Value                   |
|-------------------------------------------------|-------------------------|
| Patients analysed                               | 735 (of 742)            |
| Age, mean (SD), range                           | 28.8 (17.1), 2–83 years |
| Male                                            | 373 (50.7%)             |
| Physician diagnoses, total (median per patient) | 3,662 (5)               |
| Unanimous votes                                 | 230 (31.3%)             |
| Tied votes                                      | 100 (13.6%)             |
| Bacterial, majority vote                        | 205 (27.9%)             |
| Samsung photographs: patients (bacterial)       | 382 (57; 14.9%)         |
| Xiaomi photographs: patients (bacterial)        | 290 (127; 43.8%)        |
| Phone unknown: patients (bacterial)             | 63 (21; 33.3%)          |

### 228 4.2. Benchmark

229 Table 2 and Fig. 2 summarise the benchmark. The best model was stacked  
 230 fusion of DINOv2 image and symptom predictions (AUC 0.703, 95% CI 0.66–  
 231 0.74), followed by the frozen DINOv2 image model alone (0.700). Late, early  
 232 and gated fusion reached 0.683–0.696. Fine-tuning did not beat frozen fea-  
 233 tures: ConvNeXt-Tiny reached 0.688, DenseNet121 0.672 and MobileNetV3  
 234 0.578, close to the published 0.645 and 0.554 [7]. Symptoms alone were weak  
 235 (best 0.596), and the modified McIsaac score ranked patients worse than  
 236 chance (0.437).

237 Three models that use no throat anatomy did about as well as the pub-  
 238 lished deep networks. The phone-only model reached 0.673, above the pub-  
 239 lished image baseline and above all symptom models. Eighteen global photo  
 240 statistics reached 0.646, and the original image size 0.562.

Table 2: Discrimination under  $5 \times 5$  cross-validation (fine-tuned CNNs:  $1 \times 5$ ). “Within phone” is the size-weighted mean AUC inside each phone stratum. Balanced accuracy and macro F1 use thresholds chosen by inner cross-validation.

| Model                           | AUC [95% CI]       | Within phone | Bal. acc. | F1    | Brier |
|---------------------------------|--------------------|--------------|-----------|-------|-------|
| Stacked fusion, DINOv2 + sympt. | 0.703 [0.66, 0.74] | 0.586        | 0.659     | 0.637 | 0.218 |
| Image, frozen DINOv2            | 0.700 [0.66, 0.74] | 0.586        | 0.656     | 0.630 | 0.215 |
| Gated fusion, DINOv2            | 0.691 [0.65, 0.73] | 0.584        | 0.645     | 0.613 | 0.188 |
| Image, fine-tuned ConvNeXt-Tiny | 0.688 [0.64, 0.73] | 0.605        | 0.652     | 0.615 | 0.211 |
| Gated fusion, ConvNeXt          | 0.684 [0.64, 0.73] | 0.582        | 0.624     | 0.606 | 0.190 |
| Image, frozen ConvNeXt-Tiny     | 0.681 [0.64, 0.72] | 0.561        | 0.641     | 0.605 | 0.218 |
| Phone only                      | 0.673 [0.63, 0.71] | 0.488        | 0.668     | 0.622 | 0.182 |
| Image, fine-tuned DenseNet121   | 0.672 [0.63, 0.71] | 0.556        | 0.622     | 0.562 | 0.207 |
| Photo statistics (18 features)  | 0.646 [0.60, 0.69] | 0.536        | 0.598     | 0.559 | 0.233 |
| Symptoms, perceptron            | 0.596 [0.55, 0.64] | –            | 0.578     | 0.563 | 0.236 |
| Symptoms, LightGBM              | 0.591 [0.55, 0.64] | 0.550        | 0.561     | 0.537 | 0.240 |
| Image, fine-tuned MobileNetV3   | 0.578 [0.53, 0.62] | 0.519        | 0.550     | 0.542 | 0.324 |
| Image size only                 | 0.562 [0.52, 0.61] | 0.493        | 0.536     | 0.504 | 0.247 |
| Gated fusion, shuffled images   | 0.529 [0.48, 0.58] | 0.507        | 0.524     | 0.500 | 0.211 |
| Modified McIsaac score          | 0.437 [0.39, 0.48] | –            | 0.493     | 0.245 | 0.241 |

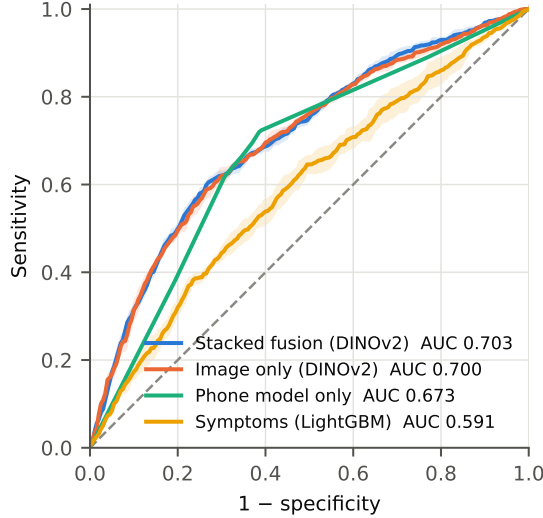


Figure 2: Mean ROC curves ( $\pm 1$  SD over five repeats) for stacked fusion (DINOv2), the frozen DINOv2 image model, the phone-only model and the LightGBM symptom model. Dashed line: chance.

#### 241 4.3. Device and site shortcuts

242 Every analysis pointed to the phone as the main source of image per-  
 243 formance (Table 3, Fig. 3). Image embeddings identified the phone almost  
 244 perfectly (AUC 0.993 for DINOv2 and 0.997 for ConvNeXt), and so did the  
 245 18 photo statistics (0.958). When AUC was computed within each phone,  
 246 image models fell from about 0.70 to 0.56–0.61, while the phone-only model  
 247 fell to 0.49, as expected. Models trained and tested inside one phone did  
 248 worse still: image AUC was 0.50 on Samsung photographs and 0.55–0.58 on  
 249 Xiaomi photographs. Models did not transfer between phones: trained on  
 250 one phone and tested on the other, image AUC was 0.50–0.53 and every 95%  
 251 confidence interval included 0.5. Restricting to photographs with an original  
 252 short side of at least 500 pixels did not change within-phone AUC (DINOv2:  
 253 0.568 Samsung, 0.611 Xiaomi), so very small crops do not explain the effect.

254 The frozen DINOv2 image model beat the phone-only model by 0.025  
 255 AUC (corrected  $t$ -test  $p = 0.035$ , uncorrected), whereas ConvNeXt did not  
 256 ( $+0.006$ ,  $p = 0.35$ ). Within the Xiaomi subgroup the photographs therefore  
 257 carry a modest signal (AUC about 0.58–0.61); within the Samsung subgroup  
 258 they carry essentially none.

Table 3: Shortcut analyses (AUC). In the cross-phone rows, the column names the phone used for testing; the model was trained on the other phone.

| Analysis                                      | Samsung            | Xiaomi             | Overall       |
|-----------------------------------------------|--------------------|--------------------|---------------|
| Identify phone: DINOv2 / ConvNeXt embeddings  | –                  | –                  | 0.993 / 0.997 |
| Identify phone: photo statistics / image size | –                  | –                  | 0.958 / 0.754 |
| Within-phone training: DINOv2 image           | 0.502              | 0.575              | –             |
| Within-phone training: ConvNeXt image         | 0.499              | 0.548              | –             |
| Within-phone training: DINOv2 + symptoms      | 0.546              | 0.561              | –             |
| Within-phone training: symptoms               | 0.627              | 0.517              | –             |
| Within-phone training: photo statistics       | 0.507              | 0.538              | –             |
| Cross-phone: DINOv2 image [95% CI]            | 0.502 [0.41, 0.59] | 0.519 [0.45, 0.58] | –             |
| Cross-phone: ConvNeXt image [95% CI]          | 0.507 [0.43, 0.59] | 0.530 [0.46, 0.60] | –             |
| Cross-phone: symptoms [95% CI]                | 0.424 [0.34, 0.52] | 0.482 [0.41, 0.55] | –             |

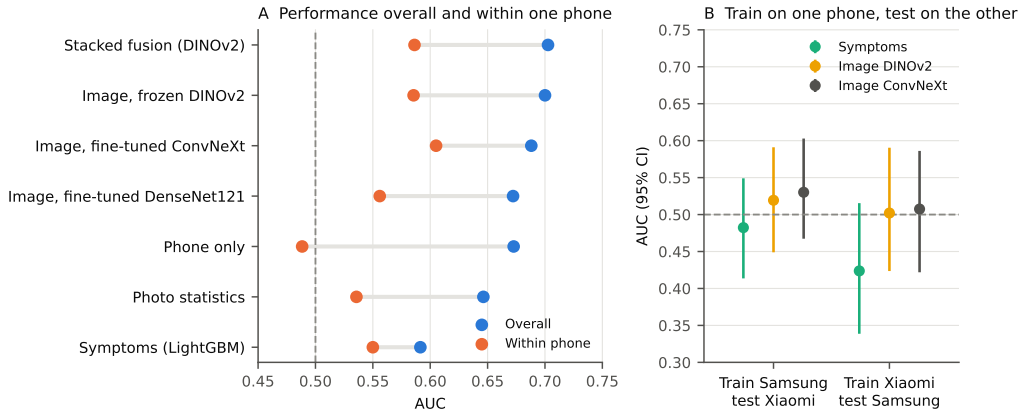


Figure 3: Device and site shortcut. (A) AUC over all patients and the size-weighted AUC within each phone; the phone-only model drops to chance within a phone. (B) Models trained on all photographs from one phone and tested on the other, with 95% bootstrap confidence intervals; all intervals include 0.5.

259 *4.4. Does fusion help?*

260 Gated fusion (DINOv2) beat the best symptom model by 0.095 AUC  
 261 (Holm-adjusted  $p = 0.012$ ; significant in 5 of 5 DeLong repeats) but did not  
 262 beat the best image-only model ( $-0.020$ ,  $p = 1.0$ ) or the best standard fusion  
 263 ( $-0.008$ ,  $p = 1.0$ ; Table 4). Stacking added 0.003 AUC over the image alone  
 264 ( $p = 0.29$ ). Within a phone, adding symptoms to the image changed AUC  
 265 by  $-0.01$  to  $+0.04$ . Symptoms therefore add little once the photograph is  
 266 known.

Table 4: Planned comparisons for gated fusion (DINOv2) under three tie rules.  $\Delta$ AUC is the mean fold-level difference;  $p$  values are one-sided, Nadeau–Bengio corrected and Holm-adjusted.

| Comparison                         | Ties bacterial           | Ties non-bacterial      | Ties dropped            |
|------------------------------------|--------------------------|-------------------------|-------------------------|
| H1: vs best symptom-only           | $+0.095$ ( $p = 0.012$ ) | $+0.044$ ( $p = 0.38$ ) | $+0.051$ ( $p = 0.22$ ) |
| H2: vs best image-only             | $-0.020$ ( $p = 1.0$ )   | $-0.009$ ( $p = 1.0$ )  | $-0.002$ ( $p = 1.0$ )  |
| H3: vs best standard fusion        | $-0.008$ ( $p = 1.0$ )   | $-0.018$ ( $p = 1.0$ )  | $-0.019$ ( $p = 1.0$ )  |
| Image (DINOv2): AUC / within phone | 0.700 / 0.586            | 0.671 / 0.609           | 0.696 / 0.605           |
| Phone only: AUC                    | 0.673                    | 0.618                   | 0.650                   |
| Best symptom-only: AUC             | 0.596                    | 0.616                   | 0.639                   |

267 *4.5. Ablations*

268 No component of the fusion network made a measurable difference (Fig. 4).  
 269 With ConvNeXt features, the full gated model reached 0.684; replacing the  
 270 gate by concatenation gave 0.686 ( $p = 0.60$ ), FiLM 0.681, removing the aux-  
 271 iliary heads 0.682 and removing modality dropout 0.682. Training on hard  
 272 instead of soft labels lowered AUC to 0.672 (difference 0.012,  $p = 0.19$ ).  
 273 Removing fever and cough did not hurt (0.690). The shuffled-image control  
 274 fell to 0.529, below symptoms alone, confirming that the pipeline does not  
 275 leak labels and that the fusion network relied almost entirely on the image  
 276 branch.

277 *4.6. Label dependence and sensitivity*

278 Models discriminated better on unanimous than on split votes (DINOv2  
 279 image: 0.719 vs 0.678; symptoms: 0.623 vs 0.580). Image predictions cor-

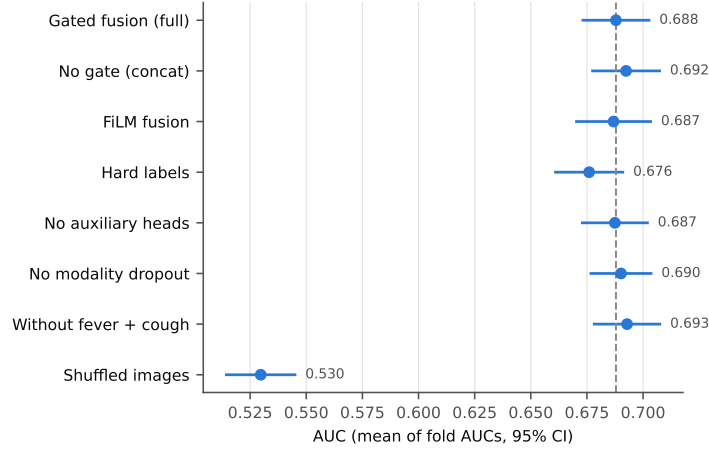


Figure 4: Ablation of the gated fusion network (ConvNeXt features): mean fold AUC with 95% confidence interval. The shuffled-image control falls to chance.

280 related more with the physicians’ vote share than symptom predictions did  
 281 (Spearman 0.32 vs 0.13), though the phone-only model also reached 0.28.  
 282 Image embeddings predicted the out-of-fold errors of the symptom model  
 283 (cross-validated  $r = 0.29$ ; permutation  $p = 0.002$ ), but this test cannot sepa-  
 284 rate throat appearance from phone identity. Under both alternative tie rules  
 285 the conclusions held: image models stayed near 0.67–0.70 overall and about  
 286 0.61 within phone, fusion never beat the image alone, and the advantage of  
 287 fusion over symptoms was no longer significant (Table 4).

#### 288 4.7. Calibration and clinical utility

289 Brier scores ranged from 0.182 (phone only) to 0.218 (stacking). Class-  
 290 balanced training made image and fusion models overestimate risk, and  
 291 decision-curve analysis showed no net benefit over treating everyone at thresh-  
 292 olds up to 0.20; at 0.30 the stacked model gave a net benefit of 0.024 versus  
 293  $-0.030$  for treat-all, but the phone-only model gave 0.082 (Fig. 5). The  
 294 models are therefore not ready for treatment decisions.

#### 295 4.8. Explainability and subgroups

296 In the symptom model, cough, age, vomiting, myalgia and sore throat  
 297 had the largest mean absolute SHAP values (Fig. A.8). The gate weighted  
 298 the image at  $g \approx 0.48$  on average (DINOv2) and slightly less when physicians

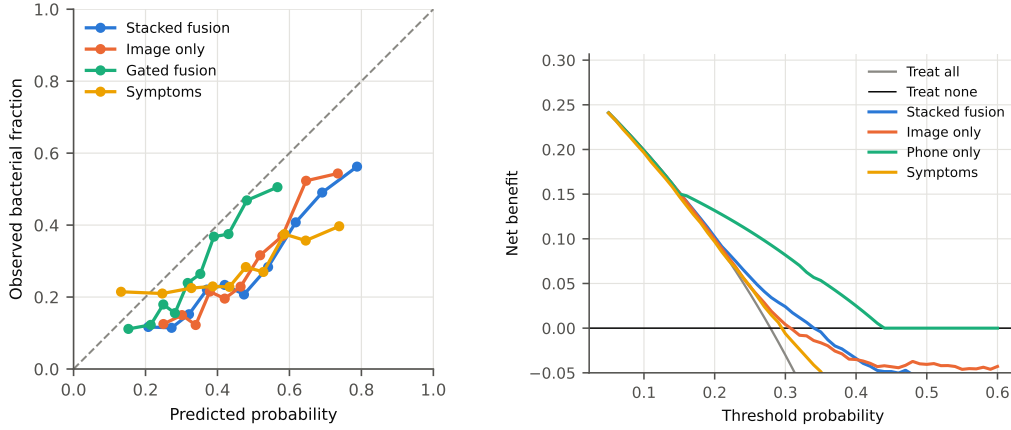


Figure 5: (A) Calibration curves and (B) decision curves for the main models. Treat-all and treat-none are shown for reference.

disagreed (median 0.47 vs 0.50, Mann–Whitney  $p < 0.001$ ; Fig. A.9). Grad-CAM maps (Fig. 6) of the most confident bacterial cases highlighted white spots in close-up photographs, whereas the most confident non-bacterial cases were wide views of the oral cavity with the uvula and tongue visible, suggesting that framing and zoom, which may differ between sites, act as an additional cue. The image model discriminated better in children aged 14 or younger (AUC 0.77) than in adults aged 45 or older (0.64), and similarly in women (0.68) and men (0.72) (Fig. A.7).

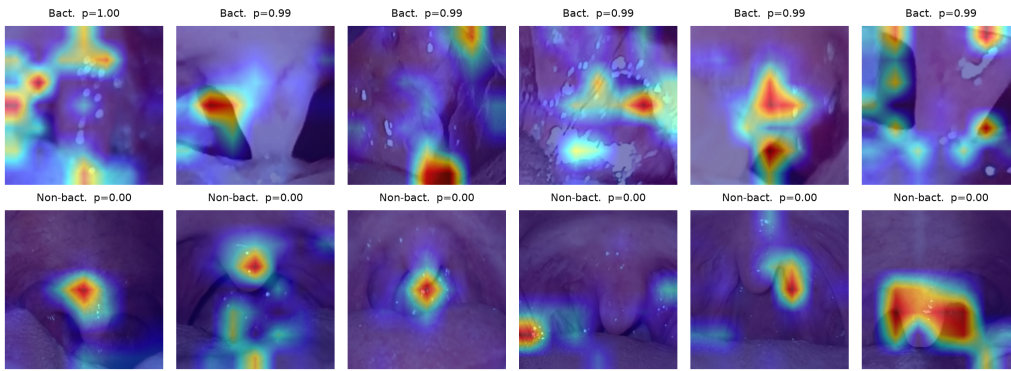


Figure 6: Grad-CAM maps of a ConvNeXt-Tiny fine-tuned on all patients for the six most confident bacterial (top) and non-bacterial (bottom) cases. Red marks high attention. Bacterial predictions are close-ups; non-bacterial predictions are wide views.

## 307 5. Discussion

308 We set out to improve smartphone pharyngitis classification by adding  
309 symptoms to throat photographs. Instead, the benchmark showed that most  
310 of the apparent image performance on PGUPharyngitis comes from recog-  
311 nising the phone, and with it the city, that produced the photograph. The  
312 phone alone predicted the expert label as well as the published deep-learning  
313 baseline, and 18 colour statistics nearly matched it. Image embeddings iden-  
314 tified the phone almost perfectly; within one phone, image models lost most  
315 of their discrimination; and no model transferred between phones.

316 *Why the shortcut exists.* Bacterial prevalence was 44% among Xiaomi pho-  
317 tographs and 15% among Samsung photographs. Whether this reflects true  
318 epidemiological differences between the two cities, different referral patterns  
319 or different labelling physicians cannot be resolved from the released data.  
320 Either way, any feature that identifies the phone (colour rendering, sharp-  
321 ness, lighting, framing) becomes predictive of the label. Our Grad-CAM  
322 maps suggest that framing differs too: confident bacterial predictions were  
323 close-ups, confident non-bacterial predictions wide views. This is the same  
324 failure mode reported for chest radiographs, where models recognised hospi-  
325 tals and scanners [10, 11, 12], and it is invisible in pooled cross-validation.

326 *What remains when the shortcut is removed.* Within the Xiaomi subgroup,  
327 image models kept a modest signal (AUC 0.58–0.61); within the Samsung  
328 subgroup, which had only 57 bacterial cases, they kept essentially none. The  
329 images do contain some information beyond symptoms, as the residual test  
330 showed, but a within-site AUC around 0.6 is far from clinically useful. The  
331 image model also worked best in children, in whom streptococcal pharyngitis  
332 is most common, which may deserve a dedicated study.

333 *Fusion.* Fusion beat symptoms alone only under one of three equally defen-  
334 sible tie rules, and it never beat the image alone. The gate, auxiliary heads,  
335 modality dropout and FiLM conditioning made no difference, and the gated  
336 model learned to weight both branches almost equally. With symptoms this  
337 weak (AUC about 0.6) and an image signal dominated by site, there is little  
338 for a fusion architecture to exploit. For future work on this dataset, the  
339 lesson is that a better fusion architecture will not fix confounded data.

340 *Symptoms and clinical scores.* The modified McIsaac score ranked patients  
341 worse than chance. Two explanations are likely: the dataset lacks exudate  
342 and lymph-node items, and the labels come from physicians who were not  
343 using a clinical score. Cough and age were the most informative symptoms,  
344 consistent with clinical scores, but together the symptoms explained little of  
345 the physicians’ decision.

346 *Implications.* Smartphone throat-imaging studies should (i) report a device-  
347 or site-only baseline, (ii) evaluate within and across acquisition sites, (iii) bal-  
348 ance or randomise devices across sites at data collection, and (iv) standardise  
349 framing. Datasets should record site and device per image. For PGUPharyn-  
350 gitis, we recommend that future work report within-phone and cross-phone  
351 AUC alongside pooled results; our folds and out-of-fold predictions make  
352 such comparisons straightforward.

### 353 5.1. Limitations

- 354 1. Labels are physician consensus from photographs and symptoms, not  
355 throat culture or rapid antigen tests, so all results concern agreement  
356 with experts rather than infection.
- 357 2. City is not recorded; we used the EXIF phone model as a proxy for site  
358 and cannot separate phone from city effects. Sixty-three photographs  
359 had no EXIF phone.
- 360 3. One hundred patients had tied votes; conclusions about fusion versus  
361 symptoms depend on the tie rule.
- 362 4. The dataset is small (205 bacterial cases) and from one country, and  
363 there is no external test set.
- 364 5. Fine-tuned CNNs were evaluated with one repeat instead of five, so  
365 their estimates are less precise.
- 366 6. The analyses were planned but not formally pre-registered and should  
367 be regarded as exploratory.
- 368 7. Grad-CAM maps were not rated by clinicians, and framing differences  
369 were observed, not measured.

## 370 6. Conclusion

371 On the largest public smartphone pharyngitis dataset, image models ap-  
372 pear to reach an AUC near 0.70, but most of this performance comes from  
373 recognising the phone and site that took the photograph. Within one site

374 the photographs carry at most a modest signal, nothing transfers across sites,  
375 and adding symptoms through any fusion design does not help. Smartphone  
376 pharyngitis models should be judged by their within-site and cross-site per-  
377 formance, against a device-only baseline, before any clinical claim is made.

## 378 **CRedit authorship contribution statement**

379 **Sandesh Bhatta:** Conceptualization, Methodology, Software, Valid-  
380 ation, Formal analysis, Investigation, Data curation, Visualization, Writing –  
381 original draft, Writing – review & editing.

## 382 **Declaration of competing interest**

383 The author declares no known competing financial interests or personal  
384 relationships that could have appeared to influence the work reported in this  
385 paper.

## 386 **Funding**

387 This research did not receive any specific grant from funding agencies in  
388 the public, commercial, or not-for-profit sectors.

## 389 **Ethics statement**

390 This study used only the anonymised, publicly released PGUPharyngi-  
391 tis dataset. The original data collection was approved by a research ethics  
392 committee (IR.BPUMS.REC.1403.282) with written informed consent from  
393 all participants [7]. No additional ethics approval was required for this sec-  
394 ondary analysis of public data.

## 395 **Data availability**

396 PGUPharyngitis is publicly available under CC BY 4.0 at <https://doi.org/10.6084/m9.figshare.28163513>. Fold assignments and out-of-fold  
397 predictions are released with the code.  
398

### 399 Code availability

400 All code, configuration files, fold assignments and out-of-fold predictions  
401 are available at <https://github.com/<username>/pharyngitis-fusion>  
402 (archived at Zenodo, DOI to be added). Two scripts reproduce every table  
403 and figure.

### 404 Declaration of generative AI and AI-assisted technologies in the 405 writing process

406 During the preparation of this work the author used Claude (Anthropic)  
407 to assist with code development, data analysis and drafting of the manuscript.  
408 After using this tool, the author reviewed and edited the content as needed  
409 and takes full responsibility for the content of the publication.

### 410 Acknowledgements

411 The author thanks the creators of PGUPharyngitis for releasing their  
412 data publicly.

### 413 Appendix A. Supplementary figures

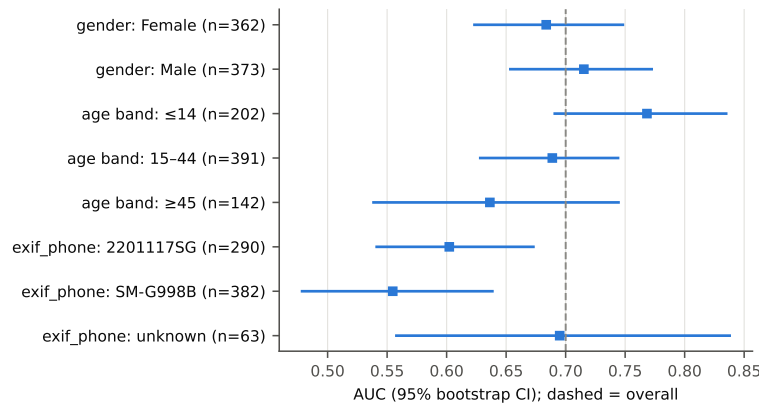


Figure A.7: AUC of the frozen DINOv2 image model by sex, age band and phone, with 95% bootstrap confidence intervals. Dashed line: overall AUC.

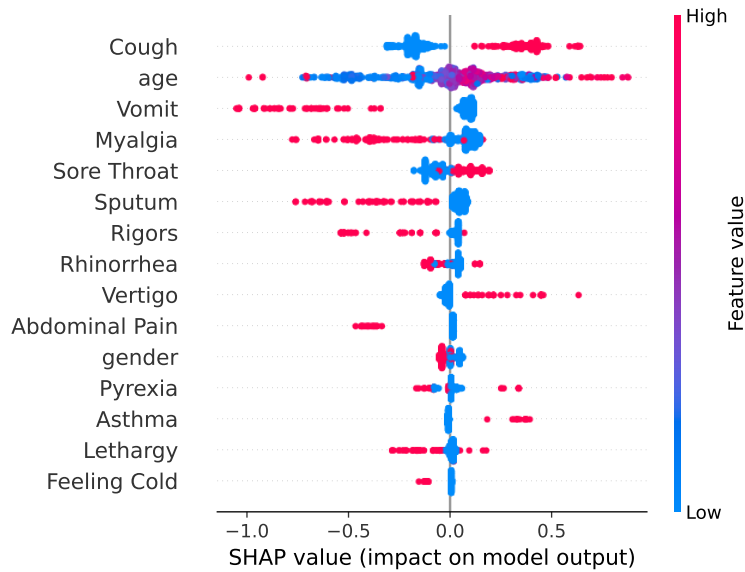


Figure A.8: SHAP summary of the LightGBM symptom model fitted on all patients.

## References

- [1] S. T. Shulman, A. L. Bisno, H. W. Clegg, M. A. Gerber, E. L. Kaplan, G. Lee, J. M. Martin, C. Van Beneden, Clinical practice guideline for the diagnosis and management of group A streptococcal pharyngitis: 2012 update by the Infectious Diseases Society of America, *Clinical Infectious Diseases* 55 (10) (2012) e86–e102. doi:10.1093/cid/cis629.
- [2] A. Spinks, P. P. Glasziou, C. B. Del Mar, Antibiotics for treatment of sore throat in children and adults, *Cochrane Database of Systematic Reviews* 12 (2021) CD000023. doi:10.1002/14651858.CD000023.pub5.
- [3] R. M. Centor, J. M. Witherspoon, H. P. Dalton, C. E. Brody, K. Link, The diagnosis of strep throat in adults in the emergency room, *Medical Decision Making* 1 (3) (1981) 239–246. doi:10.1177/0272989X8100100304.
- [4] W. J. McIsaac, D. White, D. Tannenbaum, D. E. Low, A clinical score to reduce unnecessary antibiotic use in patients with sore throat, *Canadian Medical Association Journal* 158 (1) (1998) 75–83.

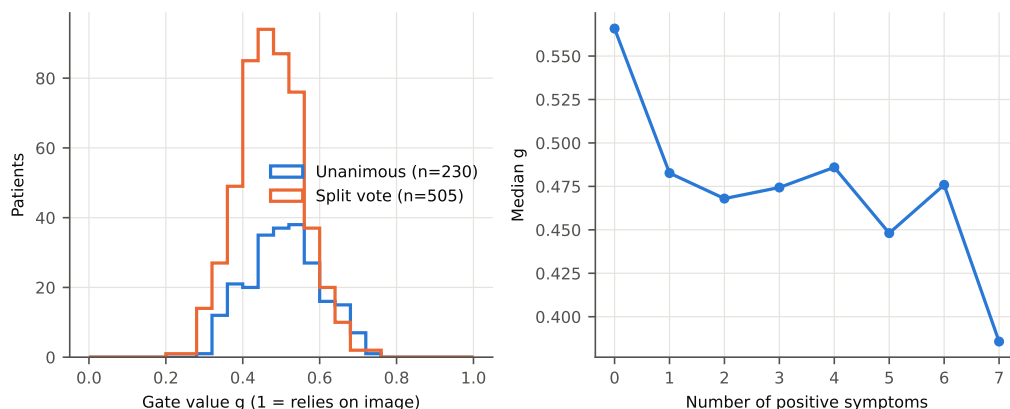


Figure A.9: Gate values of the gated fusion model (DINOv2): distribution for unanimous and split votes (left) and median gate value by number of positive symptoms (right).  $g = 1$  means the model relies only on the image.

- 431 [5] T. K. Yoo, J. Y. Choi, Y. Jang, E. Oh, I. H. Ryu, Toward automated  
432 severe pharyngitis detection with smartphone camera using deep learn-  
433 ing networks, *Computers in Biology and Medicine* 125 (2020) 103980.  
434 doi:10.1016/j.combiomed.2020.103980.
- 435 [6] B. Askarian, S. C. Yoo, J. W. Chong, Novel image processing method  
436 for detecting strep throat (streptococcal pharyngitis) using smartphone,  
437 *Sensors* 19 (15) (2019) 3307. doi:10.3390/s19153307.
- 438 [7] N. Shojaei, H. Rostami, M. Barzegar, et al., A publicly available pharyn-  
439 gitis dataset and baseline evaluations for bacterial or nonbacterial clas-  
440 sification, *Scientific Data* 12, vVERIFY: add article number (2025).  
441 doi:10.1038/s41597-025-05780-5.
- 442 [8] S. C. Huang, A. Pareek, S. Seyyedi, I. Banerjee, M. P. Lungren, Fusion  
443 of medical imaging and electronic health records using deep learning: a  
444 systematic review and implementation guidelines, *npj Digital Medicine*  
445 3 (2020) 136. doi:10.1038/s41746-020-00341-z.
- 446 [9] R. Geirhos, J.-H. Jacobsen, C. Michaelis, et al., Shortcut learning in  
447 deep neural networks, *Nature Machine Intelligence* 2 (2020) 665–673.  
448 doi:10.1038/s42256-020-00257-z.

- 449 [10] J. R. Zech, M. A. Badgeley, M. Liu, A. B. Costa, J. J. Titano, E. K.  
450 Oermann, Variable generalization performance of a deep learning model  
451 to detect pneumonia in chest radiographs: a cross-sectional study, *PLoS*  
452 *Medicine* 15 (11) (2018) e1002683. doi:10.1371/journal.pmed.10026  
453 83.
- 454 [11] A. J. DeGrave, J. D. Janizek, S.-I. Lee, AI for radiographic COVID-19  
455 detection selects shortcuts over signal, *Nature Machine Intelligence* 3  
456 (2021) 610–619, vERIFY.
- 457 [12] M. A. Badgeley, J. R. Zech, L. Oakden-Rayner, B. S. Glicksberg, M. Liu,  
458 W. Gale, M. V. McConnell, et al., Deep learning predicts hip fracture  
459 using confounding patient and healthcare variables, *npj Digital Medicine*  
460 2 (2019) 31. doi:10.1038/s41746-019-0105-1.
- 461 [13] J. Arevalo, T. Solorio, M. Montes-y Gómez, F. A. González, Gated  
462 multimodal units for information fusion, in: *International Conference on*  
463 *Learning Representations (Workshop Track)*, 2017, arXiv:1702.01992.  
464 VERIFY.
- 465 [14] E. Perez, F. Strub, H. de Vries, et al., FiLM: visual reasoning with a  
466 general conditioning layer, in: *Proceedings of the AAAI Conference on*  
467 *Artificial Intelligence*, Vol. 32, 2018. doi:10.1609/aaai.v32i1.11671.
- 468 [15] L. Oakden-Rayner, J. Dunnmon, G. Carneiro, et al., Hidden stratifica-  
469 tion causes clinically meaningful failures in machine learning for medical  
470 imaging, in: *Proceedings of the ACM Conference on Health, Inference,*  
471 *and Learning*, 2020, pp. 151–159. doi:10.1145/3368555.3384468.
- 472 [16] G. Varoquaux, V. Cheplygina, Machine learning for medical imaging:  
473 methodological failures and recommendations for the future, *npj Digital*  
474 *Medicine* 5 (2022) 48. doi:10.1038/s41746-022-00592-y.
- 475 [17] G. S. Collins, K. G. M. Moons, P. Dhiman, R. D. Riley, A. L. Beam,  
476 B. Van Calster, M. Ghassemi, et al., TRIPOD+AI statement: updated  
477 guidance for reporting clinical prediction models that use regression or  
478 machine learning methods, *BMJ* 385 (2024) e078378. doi:10.1136/bm  
479 j-2023-078378.
- 480 [18] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, T.-  
481 Y. Liu, LightGBM: a highly efficient gradient boosting decision tree,

- 482 in: Advances in Neural Information Processing Systems, Vol. 30, 2017,  
483 vERIFY.
- 484 [19] M. Oquab, T. Darcet, T. Moutakanni, et al., DINOv2: learning robust  
485 visual features without supervision, Transactions on Machine Learning  
486 ResearchArXiv:2304.07193. VERIFY (2024).
- 487 [20] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, S. Xie, A Con-  
488 vNet for the 2020s, in: Proceedings of the IEEE/CVF Conference on  
489 Computer Vision and Pattern Recognition, 2022, pp. 11976–11986, vER-  
490 IFY.
- 491 [21] G. Huang, Z. Liu, L. van der Maaten, K. Q. Weinberger, Densely con-  
492 nected convolutional networks, in: Proceedings of the IEEE Conference  
493 on Computer Vision and Pattern Recognition, 2017, pp. 2261–2269.  
494 doi:10.1109/CVPR.2017.243.
- 495 [22] A. Howard, M. Sandler, B. Chen, et al., Searching for MobileNetV3, in:  
496 Proceedings of the IEEE/CVF International Conference on Computer  
497 Vision, 2019, pp. 1314–1324. doi:10.1109/ICCV.2019.00140.
- 498 [23] C. Nadeau, Y. Bengio, Inference for the generalization error, Machine  
499 Learning 52 (2003) 239–281, vERIFY.
- 500 [24] E. R. DeLong, D. M. DeLong, D. L. Clarke-Pearson, Comparing the ar-  
501 eas under two or more correlated receiver operating characteristic curves:  
502 a nonparametric approach, Biometrics 44 (3) (1988) 837–845, vERIFY.
- 503 [25] X. Sun, W. Xu, Fast implementation of DeLong’s algorithm for compar-  
504 ing the areas under correlated receiver operating characteristic curves,  
505 IEEE Signal Processing Letters 21 (11) (2014) 1389–1393. doi:10.110  
506 9/LSP.2014.2337313.
- 507 [26] S. Holm, A simple sequentially rejective multiple test procedure, Scan-  
508 dinavian Journal of Statistics 6 (2) (1979) 65–70, vERIFY.
- 509 [27] A. J. Vickers, E. B. Elkin, Decision curve analysis: a novel method for  
510 evaluating prediction models, Medical Decision Making 26 (6) (2006)  
511 565–574. doi:10.1177/0272989X06295361.

- 512 [28] S. M. Lundberg, S.-I. Lee, A unified approach to interpreting model  
513 predictions, in: Advances in Neural Information Processing Systems,  
514 Vol. 30, 2017, vERIFY.
- 515 [29] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, D. Batra,  
516 Grad-CAM: visual explanations from deep networks via gradient-based  
517 localization, in: Proceedings of the IEEE International Conference on  
518 Computer Vision, 2017, pp. 618–626, vERIFY.